Government Office with Multiple Departments

Submitted By

Student Name	Student ID
Md Arham Tabib	221-15-5707
Sultanus Salehin	221-15-5357
Sumyia Sabrin Liza	221-15-5768
Swarnaly Sarker	221-15-5943
Md Tazrian Alam	221-15-5866

Computer Networking LAB PROJECT REPORT

This Report Presented in Partial Fulfillment of the course **CSE314: Computer Network Lab in the Computer Science and Engineering Department**

DAFFODIL INTERNATIONAL UNIVERSITY

Dhaka, Bangladesh

December 16, 2024

DECLARATION

We hereby declare that this lab project has been done by us under the supervision of Abdullah Al-Amin ,Lecturer, Department of Computer Science and Engineering, Daffodil International University. We also declare that neither this project nor any part of this project has been submitted elsewhere as lab projects.

Submitted To:

Abdullah Al-Amin
Lecturer
Department of Computer Science and Engineering Daffodil
International University

Submitted by

<table>
<tr><td colspan="2">Md Arham Tabib
ID:221-15-5768
Dept. of CSE, DIU</td></tr>
<tr><td>Sultanus Salehin
ID:221-15-5357
Dept. of CSE, DIU</td><td>Sumyia Sabrin Liza
ID:221-15-5768
Dept. of CSE, DIU</td></tr>
<tr><td>Swarnaly Sarker
ID:221-15-5943
Dept. of CSE, DIU</td><td>Md Tazrian Alam
ID:221-15-5866
Dept. of CSE, DIU</td></tr>
</table>

COURSE & PROGRAM OUTCOME

The following course has course outcomes as follows:.

Table 1: Course Outcome Statements

CO's	Statements
CO1	Define and Relate all the device and their interfaces, IP addresses, Subnet mask, default gateway
CO2	Apply Routing algorithm based on the floor/ LAN design
CO3	Create an edge router and add a server. Make the server DHCP, SMTP, HTTP, and DNS capable. Make the server reachable from any device of the topology.
CO4	Develop a topology where everything is reachable one way or another.

Table 2: Mapping of CO, PO, Blooms, KP and CEP

CO	PO	Blooms	KP	CEP
CO1	PO1	C1, C2	KP3	EP1, EP3
CO2	PO2	C2	KP3	EP1, EP3
CO3	PO3	C4, A1	KP3	EP1, EP2
CO4	PO3	C3, C6, A3, P3	KP4	EP1, EP3

The mapping justification of this table is provided in section **4.3.1**, **4.3.2** and **4.3.3**.

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Chapter 1

Introduction

The introduction chapter holds the overview of our project where we can have a brief idea about the whole project. Also the purpose of the project and the objectives behind this project.

1.1 Introduction

Technology has become an integral part of almost every aspect of organizational functioning due to the extended functions of organizational operations by the digital era. In the wake of new trends in networking and communication technology, organisations have no option than to apply appropriate modern mechanisms catering for productivity, connection, and scalability. One of the approaches that help to exclude the main disadvantages of the conventional wired networks is the implementation of wireless networking.

Specifically, this project, which is a Government office having a number of Departments, will endeavour towards portraying and implementing a network for Multiple-departmental organization with Wired as well as Wireless connectivity system. All departments operate in their own Virtual Local Area Network (VLAN), to deliver resource optimization, strict security, and bandwidth partitioning. Another important factor through which organizations ensure that the workplace is more interesting and productive is through use of wireless networking, which allows the various staff members in an organization to easily communicate and access various resources within the organization without necessarily having to be physically close.

They also have the basic supporting services including DNS, email and HTTP server to ensure that all the users in different departments are provided with the basic resources. This research shows an example of how departments might turn to recognising the digital era, use the wireless technologies and solve such questions as scalability, mobility and efficient resource sharing by proving the wired and wireless networks interacting with each other for the purpose of the organization.

1.2 Motivation

An effective and scalable network is required for the trading floor support center's growth in order to accommodate many workers in various departments. For daily operations to run smoothly, a dependable network is essential, especially for applications like data transmission, communication tools, and remote access. Furthermore, the network needs to be future-proof to facilitate future expansion and seamless integration of new services. With the implementation of this new network, we hope to improve security, decrease downtime, and boost operational efficiency.

1.3 Objectives

The objectives of this project include:

- Designing a hierarchical network to ensure scalability and redundancy.
- Providing wireless connectivity in each department for ease of access and mobility.
- Implementing VLANs to segregate departments, ensuring security and better traffic management.
- Allocating IP addresses efficiently through subnetting based on departmental needs.
- Configuring DHCP servers for dynamic IP allocation.
- Ensuring security with SSH, ACLs, port security, and secure communications.
- Using OSPF as a routing protocol to provide efficient, scalable routing.
- Providing NAT overload (PAT) to enable access to external networks.

1.4 Feasibility Study

Cisco Packet Tracer simulations, which offer a dependable platform for modeling the network architecture and testing its features, are used to verify the project's viability. This method guarantees that the design satisfies industry requirements for redundancy, security, and network scalability. Due to the widespread usage of technologies like OSPF routing, VLANs, and multilayer switches in workplace settings, this project is technically and financially viable.

1. **Packet Tracer: What is it?**

Cisco Systems created Packet Tracer, a cross-platform visual simulation tool that lets users create network topologies and simulate contemporary computer networks. Users may use the program to mimic the configuration of Cisco switches and routers using a duplicated CLI. Packet Tracer's drag-and-drop user interface lets users create and delete fictitious network devices as required. Students participating in the Certified Cisco Network Associate Academy are the program's primary target audience, and it serves as a teaching tool to help them understand the core concepts of CCNA. The tool was formerly available for free download and educational usage by students enrolled in CCNA Academy classes.

The router

A router is a device that directs data packets based on their IP addresses, much like a switch. The router is mostly a network-layer device. LANs and WANs are typically connected by routers. They use a dynamically updated routing table to determine how to route data packets. The broadcast domains of hosts connecting to a router are divided.

Change

Network switches are networking devices that use packet switching to deliver and receive data between devices on a computer network. In technical terms, it is sometimes referred to as a switching hub, bridging hub, or MAC bridge. A network switch is a multiport network bridge that forwards data at the OSI model's data link layer, or layer 2, using MAC addresses. By adding the capability of routing,

Packets on the network

A structured data unit transported by a packet-switched network is called a network packet. Control information and user data, sometimes referred to as the payload, make up a packet.

1.5 Gap Analysis

The redundancy and security measures required to accommodate the increasing number of users and devices are absent from the antiquated network architecture. Additionally, the current system lacks comprehensive security setups and failover mechanisms, which might result in possible failures and data breaches. By offering strong routing, VLAN segregation, redundant hardware, and extensive security options, the project seeks to close these gaps.

1.6 Project Outcome

When the project is finished, the trading floor support center will have a fully operational, scalable, and secure network design. The network will provide redundancy at both the internal and external connection points and accommodate many workers spread over three levels. Additionally, the architecture will accommodate future expansion, making it a long-term solution that enhances operational efficiency and satisfies corporate demands.

Chapter 2

Proposed Methodology/Architecture

This section provides the requirements analysis, design specification, methodology and overall project plan of our project.

2.1 Requirement Analysis & Design Specification

1. Software and hardware requirements

- A proper workstation
- Packet tracer by Cisco
- 8 GB of RAM
- Any 10,000+ Average CPU Mark scored Processor
- 16 GB of dedicated hard disk space

Network Requirements:

- We have divided the enterprise network into two distinct areas(**Department area, Admin area**)
- Each router handles a separate area (Departments vs. Central Services or another segment).
- The two servers can host services for separate areas, ensuring optimal performance. **If one server fails, the other server can still provide some critical services, reducing downtime.**
- Each multilayer switch serves as a central point for inter-VLAN routing within its specific area. By using two multilayer switches, it can keep traffic within each area, ensuring better segmentation and easier troubleshooting.

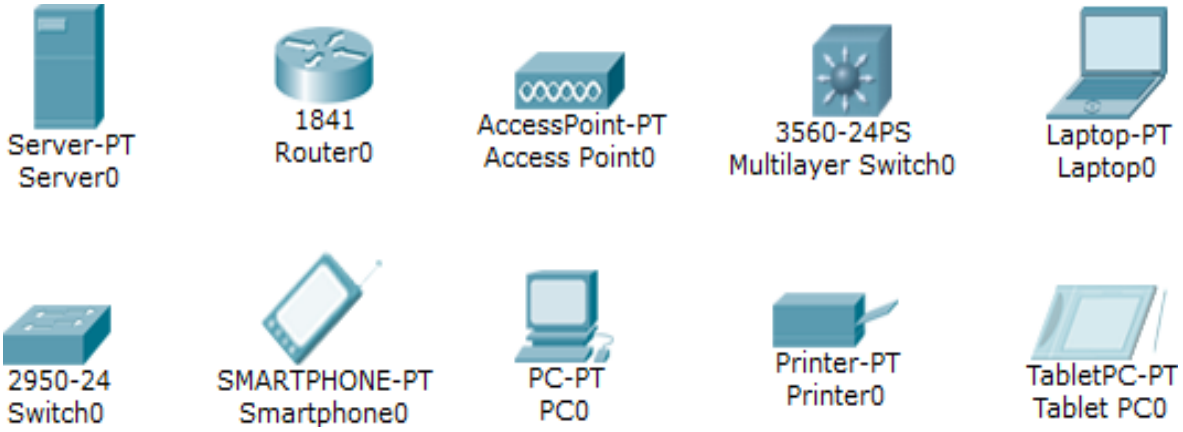
2.1.1 Overview

The overall project contains the devices used in the Network

Devices	Quantity
Router	2
Switch	7
Email server	2
DNS server	2

WEB server	2
Wireless Device(Access Point)	7
PC	14
Laptop	5
Smartphone	5
Printer	7
Multilayer Switch	2
DHCP Server	2

Devices used in this network:



2.1.2 Proposed Methodology/ System Design

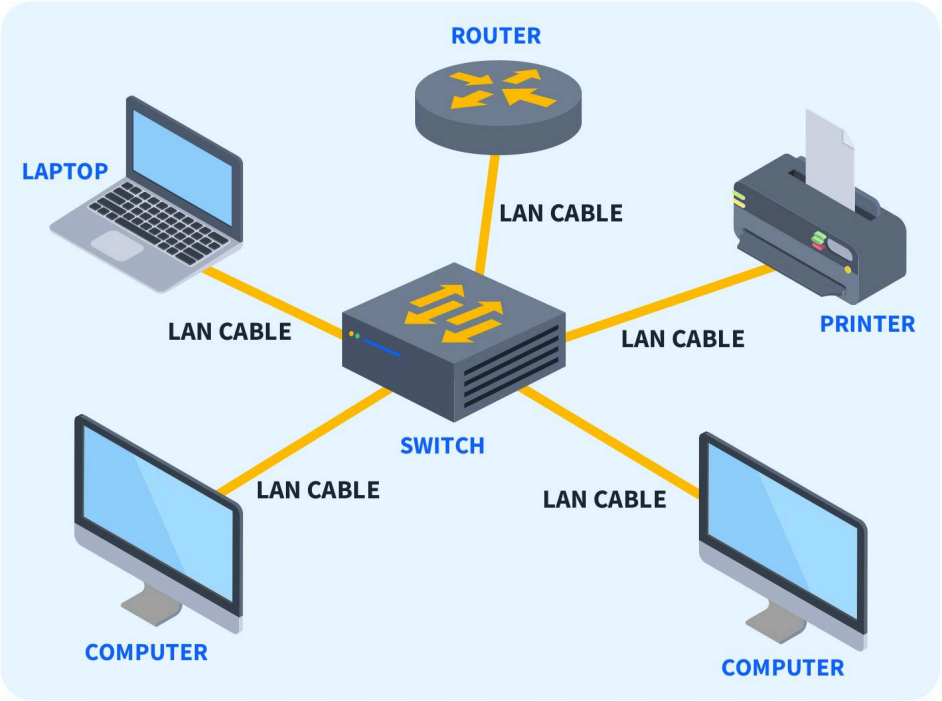
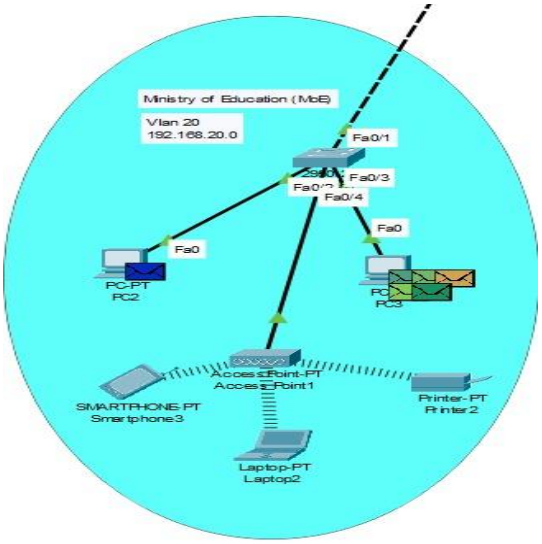


Figure 2.1: A real-life sample for each department configuration

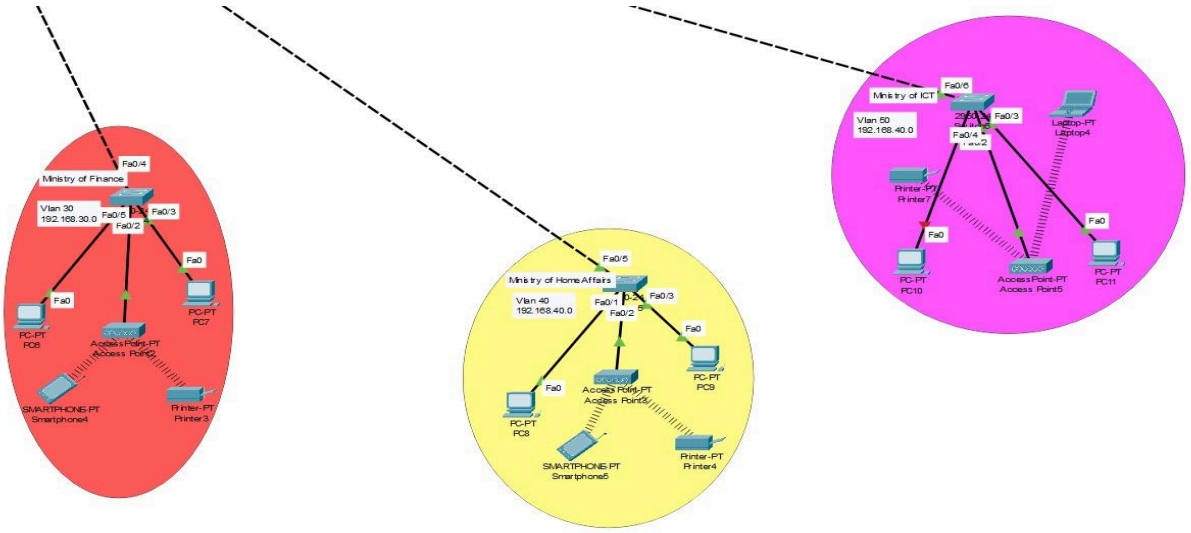
2.1.3 UI Design

The design specification for each department:

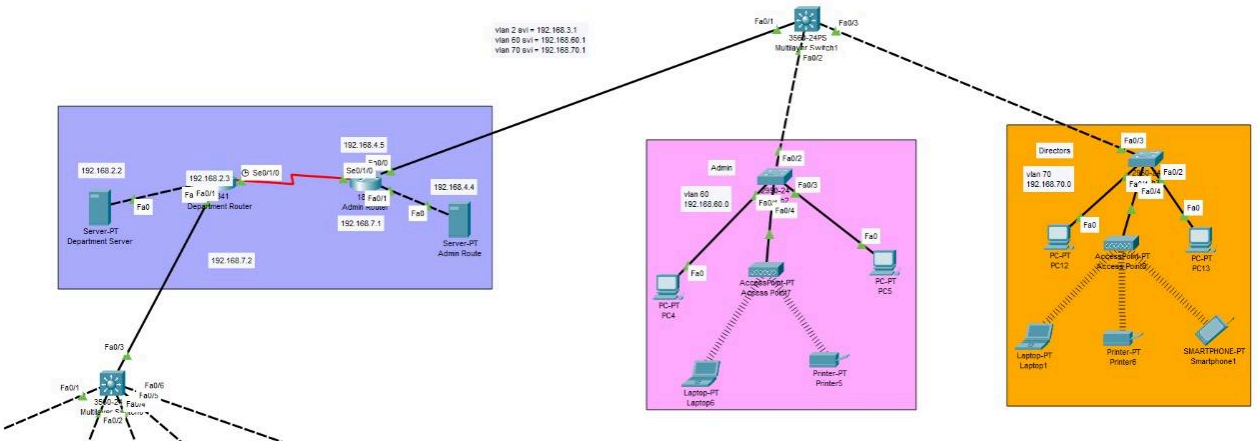
1. **Department of Education:**



2. **Department of Finance, Home affairs, ICT:**



3. **Admin directors**

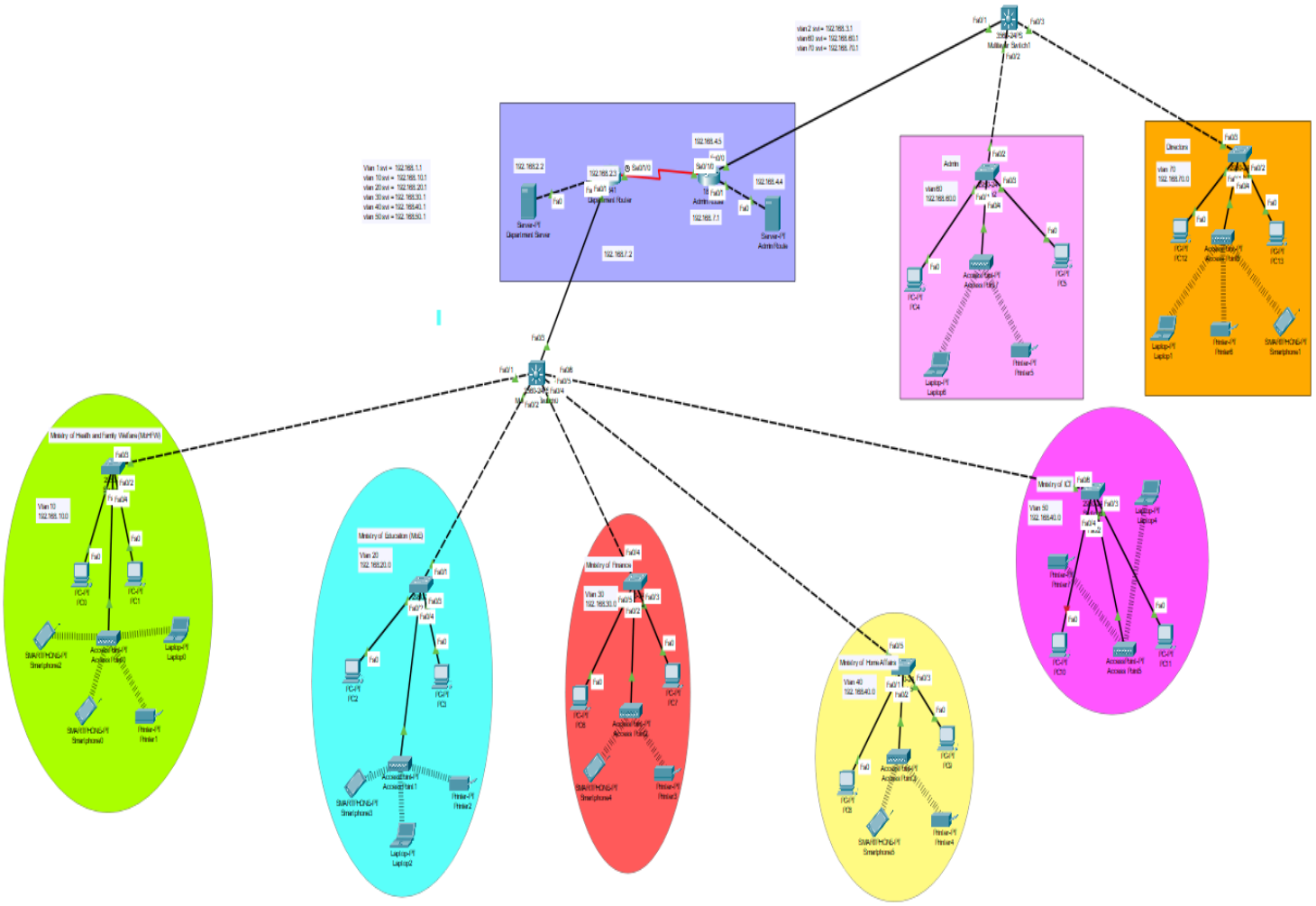


2.2 Overall Project Plan

1. Requirement Analysis: Review network needs based on the number of users and

departments.

2. Network Topology Design: Create the hierarchical design using Cisco Packet Tracer, ensuring redundancy at every layer.
3. Device Configuration: Configure routers, switches, VLANs, IP addressing, DHCP, OSPF, and security features.
4. Testing and Verification: Test the network to ensure it meets performance, redundancy, and security requirements.



Chapter 3

Implementation and Results

This section has the implementation and result part of our government office project.

3.1 Implementation

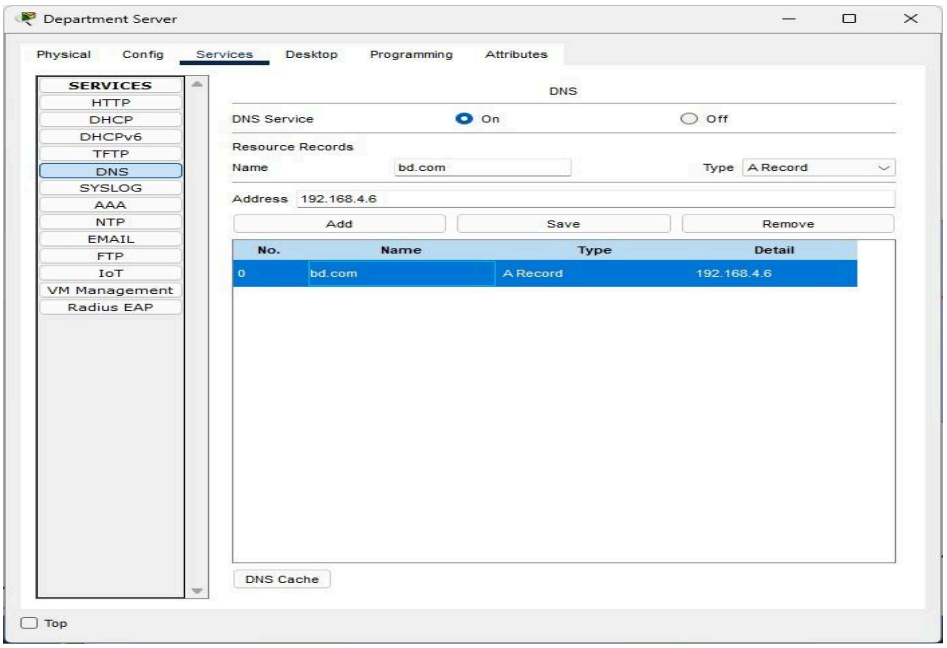
The following implementations have been done in our project.

Server

A server is a computer or system that uses a network to supply other computers, usually referred to as clients, with resources, data, services, or applications. Theoretically, a server is any computer that shares resources with a client computer. Mail servers, web servers, and virtual servers are just a few of the numerous varieties of servers. Many networks have one or more of the typical servers. Our project uses the servers listed below:

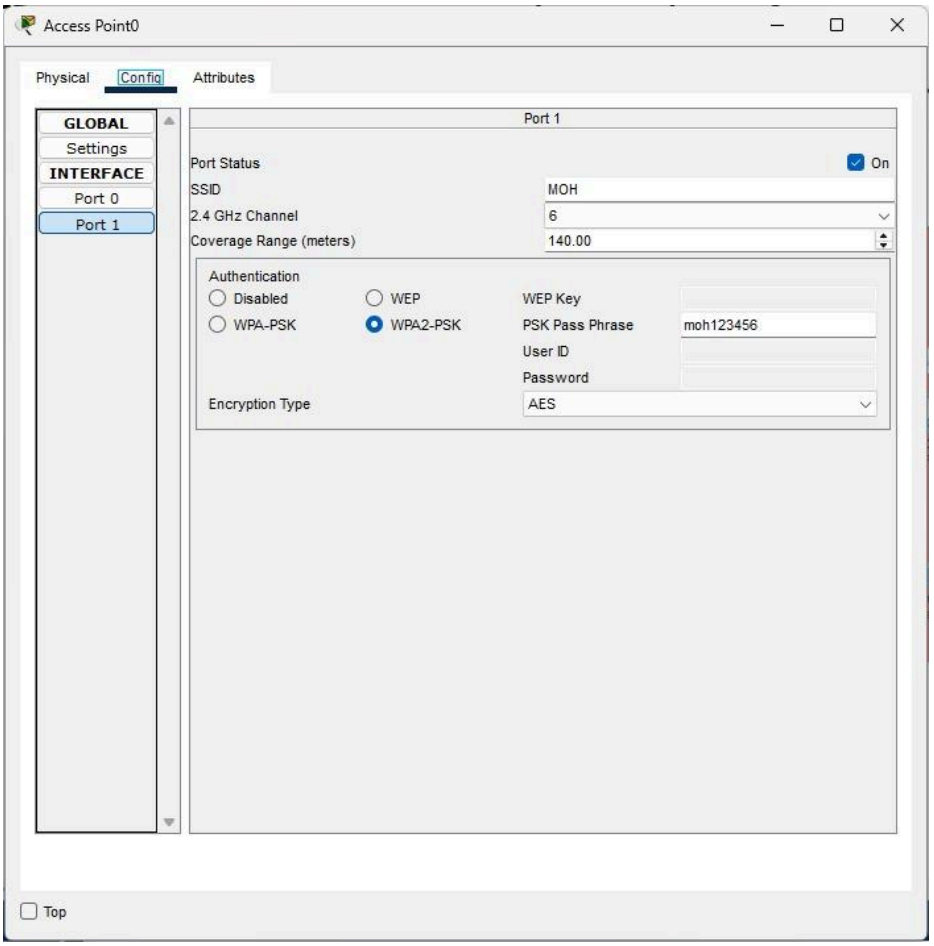
The DNS server

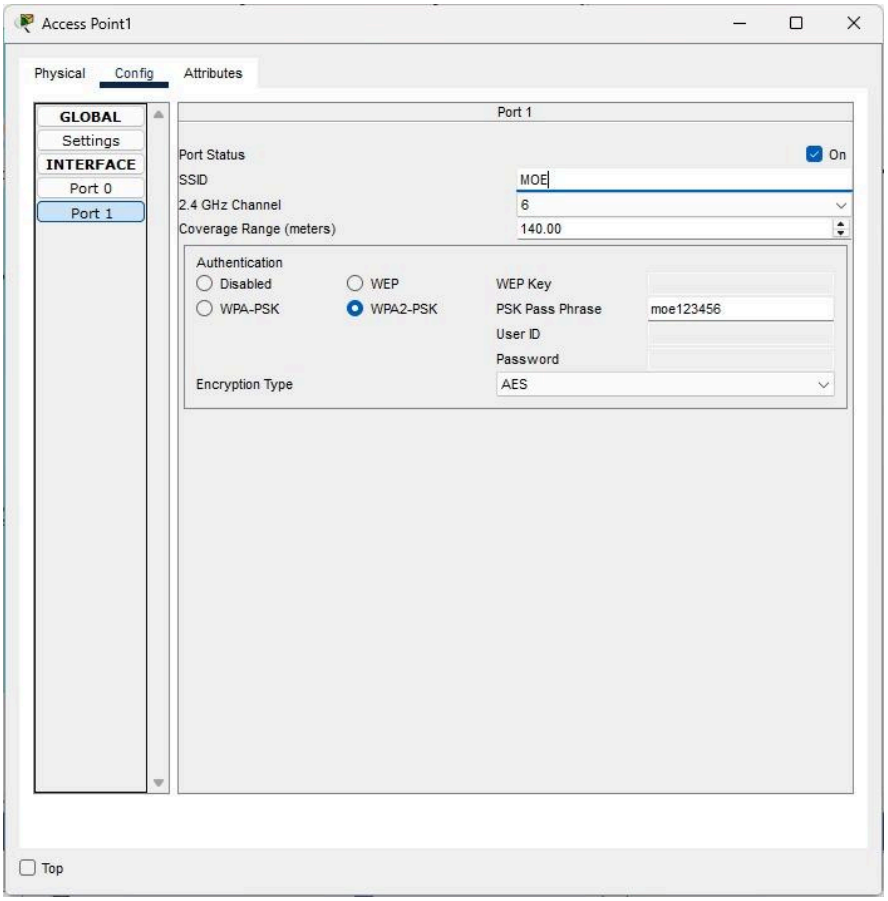
In order for consumers to comprehend IP addresses, DNS, or domain name systems, servers are application servers that give user computers a naming scheme that is easy to understand. An otherwise unknown machine name can be requested via any DNS server. A globally distributed database of names makes up the DNS system. When a user needs to know a system's address, they send a DNS request to a DNS server with the resource's name. The DNS server retrieves the necessary IP address from its database of names.

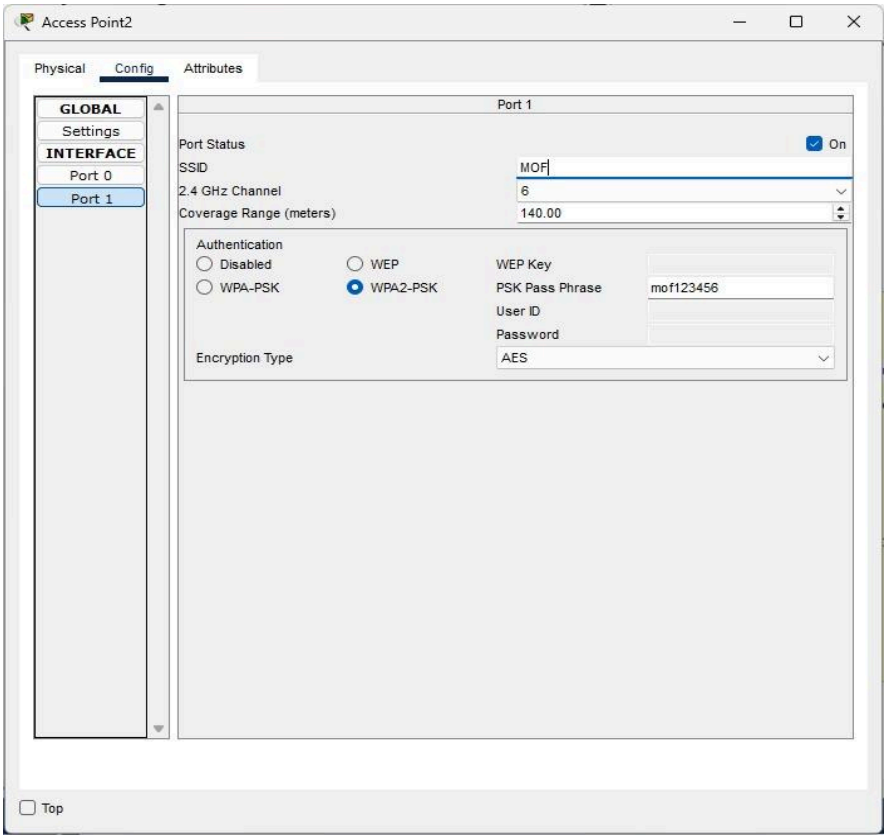


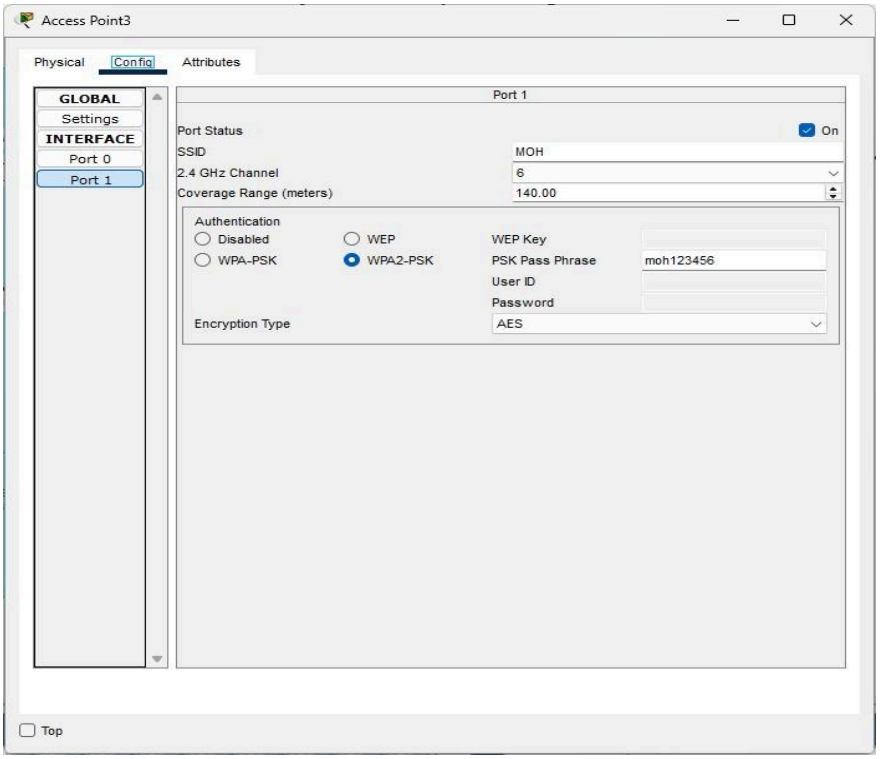
Wireless Network

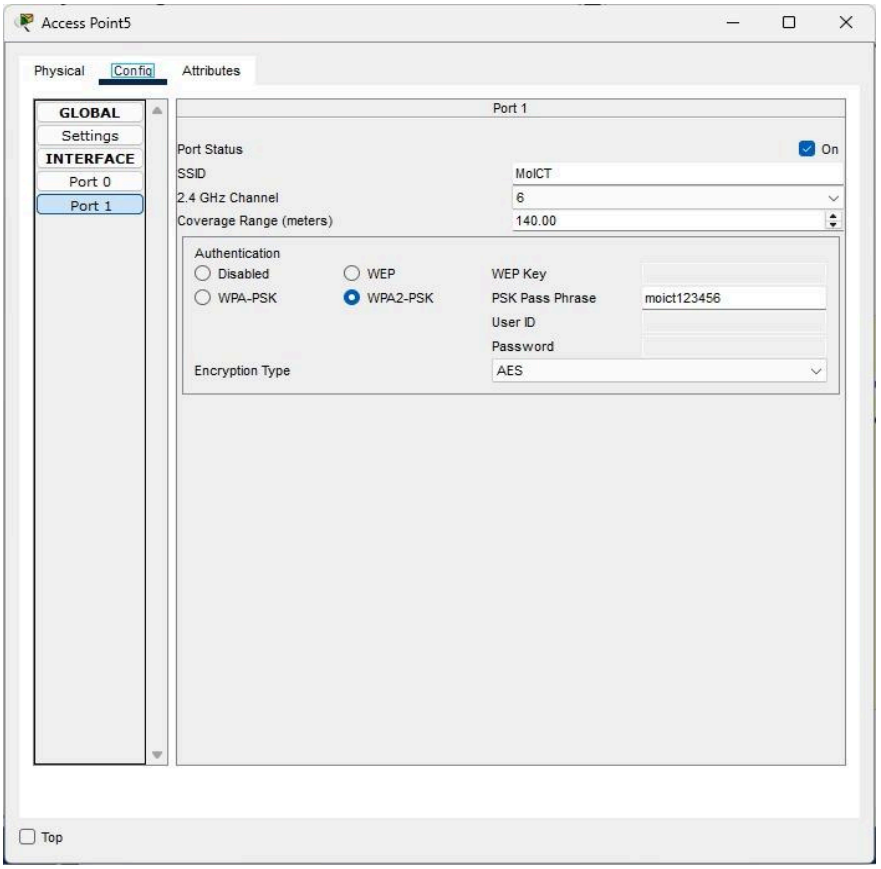
A wireless network broadcasts an access signal to the PCs or workstations. This allows PCs, laptops, and tablets to roam from room to room while maintaining a robust network connection. For wireless networks, additional security needs exist.




Access Point1
Physical
Config
Attributes
GLOBAL
Settings
INTERFACE
Port 0
Port 1
Port 1
Port Status
On
SSID
MOE
2.4 GHz Channel
6
Coverage Range (meters)
140.00
Authentication
Disabled
WEP
WEP Key
WPA-PSK
WPA2-PSK
PSK Pass Phrase
moe123456
User ID
Password
Encryption Type
AES
Top


Access Point2
Physical
Config
Attributes
GLOBAL
Settings
INTERFACE
Port 0
Port 1
Port 1
Port Status
On
SSID
MOF
2.4 GHz Channel
6
Coverage Range (meters)
140.00
Authentication
Disabled
WEP
WEP Key
WPA-PSK
WPA2-PSK
PSK Pass Phrase
mof123456
User ID
Password
Encryption Type
AES
Top


Access Point3
Physical
Config
Attributes
GLOBAL
Settings
INTERFACE
Port 0
Port 1
Port 1
Port Status
On
SSID
MOH
2.4 GHz Channel
6
Coverage Range (meters)
140.00
Authentication
Disabled
WEP
WPA-PSK
WPA2-PSK
WEP Key
PSK Pass Phrase
moh123456
User ID
Password
Encryption Type
AES
Top


Access Point5
Physical
Config
Attributes
GLOBAL
Settings
INTERFACE
Port 0
Port 1
Port 1
Port Status
On
SSID
MoICT
2.4 GHz Channel
6
Coverage Range (meters)
140.00
Authentication
Disabled
WEP
WPA-PSK
WPA2-PSK
WEP Key
PSK Pass Phrase
moict123456
User ID
Password
Encryption Type
AES
Top

Ethernet

Our network's backbone is this. It usually has a data transfer rate of 100 MB/s and is made up of cabling. It is a technique for joining several computer systems to create a local area network, complete with rules to regulate data flow and prevent simultaneous transmission from several devices. We have utilized gigabit Ethernet, one of the several forms of Ethernet that can efficiently and rapidly transfer data at 1000 Mbps. One kind of Ethernet network is Ethernet, which has a 100 Mbps data transfer rate.Electronic devices known as computing devices receive inputs from users, process them, and then output the results. These gadgets could include printers, laptops, desktop computers, smartphones, and many more.

The Internet Protocol

Internet Protocol (IP) is one of the fundamental protocols that enable the operation of the Internet. IP addresses, which are a unique set of numbers on each network, allow machines to communicate with each other via a network. It is implemented at the internet layer in the IP/TCP architecture.

VLAN (Virtual Local Area Network)

A logical grouping of devices known as a VLAN (Virtual Local Area Network) allows devices in different physical locations to interact as if they were on the same physical LAN. Because VLANs provide segmentation, improved security, and greater bandwidth utilization, they are a crucial part of business networking.

```
sw1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
sw1(config)#interface FastEthernet0/2
sw1(config-if)#exit
sw1(config)#interface vlan 20
sw1(config-if)#ip address 169.254.61.1 255.255.0.0
sw1(config-if)#exit
sw1(config)#access-list 100 deny ip 169.254.61.0 0.0.255.255 169.254.219.0 0.0.255.25
sw1(config)#access-list 100 deny ip 169.254.219.0 0.0.255.255 169.254.61.0 0.0.255.25
sw1(config)#interface vlan 20
sw1(config-if)#ip access-group 100 in
sw1(config-if)#exit
sw1(config)#interface vlan 30
sw1(config-if)#ip access-group 100 in
sw1(config-if)#exit
```

VLAN Advantages for our Project:

Network Segmentation: By splitting the network into smaller broadcast domains, VLANs improve overall performance and cut down on superfluous traffic. For instance, every department in your project—IT, sales, finance, etc.—has a unique VLAN assigned to it. By separating traffic from other departments, it enhances security and efficiency.

Enhanced Security: Sensitive data from one department is shielded from unwanted access by another by separating departments into distinct VLANs.

Simplified Network Management: By enabling logical network segmentation, VLANs facilitate traffic management and problem-solving.

Flexibility: No matter where they are physically located, devices in the same VLAN can connect with one another. This is especially helpful in settings where mobile and wireless devices are prevalent.

EIGRP (Enhanced Interior Gateway Routing Protocol)

Cisco created the dynamic routing protocol known as EIGRP. It guarantees effective and scalable routing while functioning as part of an autonomous system. Because of its flexibility, low bandwidth consumption, and quick convergence, EIGRP is frequently utilized in enterprise networks.

How EIGRP Works

Operates: Neighbor Discovery: EIGRP routers use Hello packets to create neighbor relationships in order to find and preserve connections with devices that are directly connected.
Topology Table: To facilitate quick recovery in the event of a link breakdown, routers keep track of a topology table that contains several routes to every destination.
DUAL Algorithm: To choose the quickest and most dependable route to each destination, EIGRP employs the DUAL (Diffusing Update Algorithm).

EIGRP Advantages for Our Project:

Rapid Convergence: EIGRP responds swiftly to network modifications, guaranteeing little downtime in the event that a device or link fails.

Equal-cost and unequal-cost load balancing are supported by EIGRP, which enhances bandwidth usage over several lines.
Scalability: EIGRP is perfect for a company with several departments, like our, because it can manage big networks.
Configuration Ease: EIGRP is simple to set up and manage, particularly in a setting with several VLANs that require effective routing.

Department Router Configuration

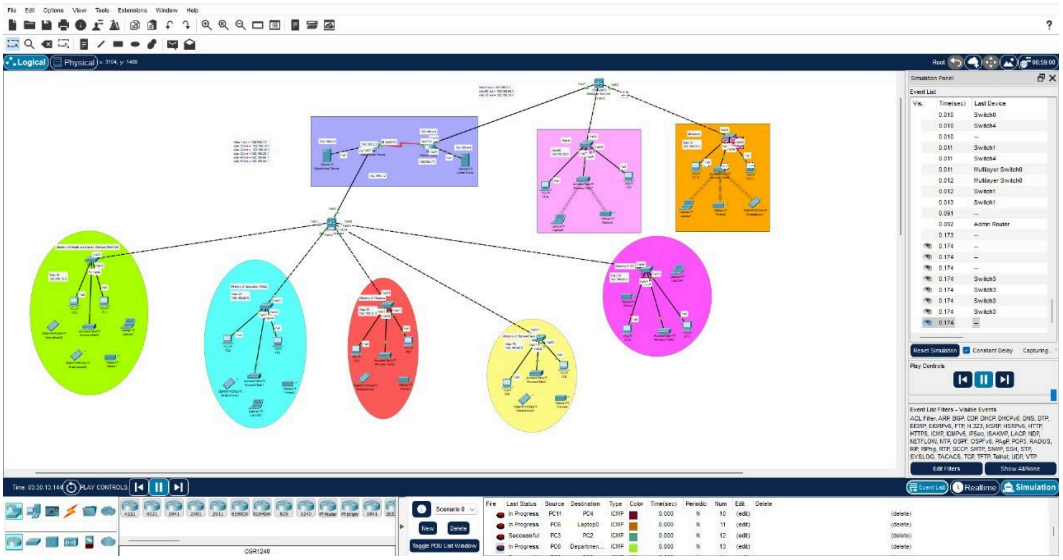
3.2 Performance Analysis

Redundancy Testing: The network was tested for failover scenarios with two ISPs and redundant routers and switches.
• DHCP Functionality: Devices successfully received dynamic IP addresses via the DHCP server.

• Security Testing: SSH and ACL configurations were verified for secure communication.

3.3 Results and Discussion

Finally, we have combined all the steps mentioned in the working process) and Implemented the desired Enterprise Network with Departments. The proposed Architecture, when simulated on Cisco Packet Tracer, Produced results which are Demonstrated as follows:



Chapter 4

Engineering Standards and Mapping

4.1 Impact on Society, Environment and Sustainability

This network project has a profound influence on society by easing the flow of information in governmental working processes and improving cooperation between departments and with citizens. In a way, it improves resource optimization, and reduces paper usage that may otherwise be physical and detrimental to the environment. The efficiency of the hybrid network decreases power consumption thus making it a future solution for government operations.

4.1.1 Impact on Life

The project has positive impacts on the quality of life for both employees and every citizen of the country. People are in a position to gain better connected networks, mobility and secure means to have access to their necessary tools and docs for the flow of work and satisfaction thus improving flow of work. This leads to enhancing the performance of users and various levels of government bodies resulting in better response and strengthened credibility.

4.1.2 Impact on Society & Environment

This way, the government departments work in a well connected network and share information and data much more easily. Through execution of remote and digital processes, it eliminates the carbon cost from manual processes and physical traveling. The network therefore encourages paperless policies thus preserving the environment and local and international environmental conservation policies.

4.1.3 Ethical Aspects

Some of the ethical issues are to do with protecting data from being accessed by unauthorized personnel, protection of communication channels and data privacy. Measures such as ACLs and SSH ensure that; the principle of confidentiality and integrity of sensitive data belonging to the government are maintained. Another couple of ethical aspects includes clarity about the operational transparency of the networks as well as conformance to the local laws.

4.1.4 Sustainability Plan

This paper defines a sustainability plan as one that aims at extending the life of the network and its performance to further years by updating the existing hardware and software, continued compliance with energy conservation policies, and growth preparedness. Seminars for staff on appropriate conduct on the networks and over time checks on security measures will ensure that the system remains functional and relevant in the ever-changing computer environment.

4.2 Project Management and Teamwork

The main strength of the project reported here is the project management and coordination that required both technological as well as structured operational skills in working with a team. The management strategy of the project was very much formal and systematic, hence the project objectives were met within the given timeline and available resources. Below is a detailed breakdown of the project management and teamwork processes:

Planning and Organization

The planning started with the setting of aims and objectives of the project as well as the understanding of its proposed coverage and expected outcome. The project schedule was created with Gantt charts to use time and work efficiently to plan and reach the goals set in the project, including designing the network architecture, configurations of VLANs, and incorporating security. Mnemonic five: accountability The assignment of roles clarifying expectations between staff members and giving each team member information about descriptions and levels of interaction.

Organization and Co-Operation

The project teams were created in the functional groups according to their specialization, the fields are: network design, configuration, test and documentation. Team meetings were held frequently in order to ensure closer cooperation, discuss different stages of the project and solve arising problems. Communications were both face to face and by electronic means through the available platforms of slack and zoom to ensure that the working members in the team are kept informed and updated.

Organizational Assignment and Responsibility

Individual assignments that include setting up DHCP server, OSPF routing and wireless connectivity were assigned to individual team members depending on their best specialty. Going this way not only made the work more effective but also ensured that all members of the teams themselves were responsible for performing certain tasks. These included project management tools like Trello where we documented the tasks which were completed and by whom to maximize accountability.

Problem-Solving and Decision-Making

This element covers problem-solving and decision making because decision making is central to problem solving.
Problems like mismatch of IP allocation or routing issues were resolved in the network. A systematic problem-solving approach was adopted: defining the problem on hand, generating possible solutions as well as executing the best solution. Group decisions were made, everything was discussed and everybody's input was valued taking into consideration group responsibility and creativity.

Final Presentation and Handover

Here, vital information will also be presented to the client, and they will be given the project's final output. As a final step in the project all the network details including its design, its functionality, and performance was presented to the stakeholders. Every member of the team defined his role in the project and the most important accomplishments of the team. For later

use, detailed documentation of the structure of the entire network as well as configurations was developed, including a user guide.

4.3 Complex Engineering Problem

4.3.1 Mapping of Program Outcome

In this section, we provided a mapping of the problem and provided a solution with targeted Program Outcomes (PO's).

Table 4.1: Justification of Program Outcomes

PO's	**Justification**
PO2	Analyze an adaptable approach to network configuration and optimization by analyzing IP address allocation, evaluating current and emerging communication protocols to configure routers, switches, and servers.
PO3	Analyze an adaptable approach to network configuration and optimization by analyzing IP address allocation, evaluating current and emerging communication protocols to configure routers, switches, and servers.
PO4	Analyze an adaptable approach to network configuration and optimization by analyzing IP address allocation, evaluating current and emerging communication protocols to configure routers, switches, and servers.

4.3.2 Complex Problem Solving

In this section, we provided a mapping with problem solving categories. For each mapping add subsections to put rationale (seeTable 4.2). For P1, you need to put another mapping with

Knowledge profile and rationale thereof.

Table 4.2: Mapping with complex problem solving.

EP1 Dept of Knowledge	**EP2** Range of Conflicting Requirements	**EP3** Depth of Analysis	**EP4** Familiarity of Issues	**EP5** Extent of Applicable Codes	**EP6** Extent Of Stakeholder Involvement	**EP7** Inter-dependence
✔	✔	✔				

4.3.3 Engineering Activities

In this section, provided a mapping with engineering activities. For each mapping add subsections to put rationale (see Table 4.3).

Table 4.3: Mapping with complex engineering activities.

EA1 Range of resources	**EA2** Level of Interaction	**EA3** Innovation	**EA4** Consequences for society and environment	**EA5** Familiarity
✔		✔		

EA1: Utilized switches, routers, wireless access points, and DHCP servers, web servers, email servers.

EA3: Integration of advanced configurations like inter-VLAN routing, OSPF, and PAT.

Chapter 5

Conclusion

This section has the overall summary, limitations and future work of our project.

5.1 Summary

This project's enterprise network design and implementation shows a strong and scalable architecture that is suited to the unique requirements of a multi-departmental company. In order to improve network performance and security, each department has been set up with its own VLAN, guaranteeing logical traffic segregation. The multi-layer switches and routers that make up the network's backbone allow for smooth interdepartmental communication while upholding stringent access controls.
In order to meet the expectations of the modern workplace, wireless access points and printers are integrated to provide end users with freedom and convenience. Deploying department-specific servers also improves data accessibility and simplifies administrative tasks, which promotes effective resource use throughout the company.

Scalability and security are given top priority in this network architecture, which enables it to be adjusted to future organizational expansion and changing technology requirements. By lowering broadcast traffic and streamlining network administration, IP subnetting and VLANs show that enterprise problems are approached with consideration.

A network system that strikes a balance between scalability, security, and efficiency is demonstrated by this project, which concludes. Establishing such a network puts the company in a strong position to meet its operational needs and take advantage of upcoming technological developments.

5.2 Limitation

The project's reliance on simulation tools, such as Cisco Packet Tracer, which could not accurately represent real-world hardware and performance settings, is one of its limitations. Furthermore, even if the network is built to scale, time, money, and physical hardware limits
Limitations may affect the amount of advanced features and redundancy that may be used. Challenges may also arise from the intricacy of integrating emerging technology and guaranteeing compatibility with current systems. Last but not least, even though the network's security measures are extensive, they might need to be modified frequently as new threats appear.

5.3 Future Work

The following project will have these updates in future:

1. Hardware Implementation
2. Cloud Integration

3. Advanced Security Protocols
4. Network Monitoring and Automation
5. IoT Integration

References

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